



Introduction to Artificial Intelligence (INTRAI)

Module 3: Machine Learning Basics

What is Machine Learning

Lesson Notes

KMVTECHUG

Version: 2026

Machine Learning

Machine learning (ML) is a way to build models by learning patterns from data instead of writing every rule by hand.

If you show many examples of healthy leaf and diseased leaf, the system can learn features that separate them and then classify a new photo. That is supervised learning in plain form.

Supervised Learning

Supervised learning uses labelled examples: input plus the correct answer. Classification puts items into categories. Regression predicts a number.

Campus example: predict whether a support ticket is about fees, exams, or housing using past labelled tickets.

Unsupervised and Reinforcement

Unsupervised learning finds structure without labels, for example grouping similar customers.

Reinforcement learning learns by trial and feedback (reward or penalty). It is used in games and some robotics.

You only need the idea of each type for this beginner course.

Reflect: Which type of learning (supervised, unsupervised, or reinforcement) best matches grading past exam papers with known correct answers?

Closing

Next lesson: training versus testing, overfitting, and bias linked to data quality.